

# **Tensor Operators for Determinantal Product States of $I^N$ Configurations**

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## Tensor operators

Definition:

$$\begin{aligned}[J_z, T_q^{(k)}] &= q T_q^{(k)} \\ [J_{\pm}, T_q^{(k)}] &= \sqrt{k(k+1) - q(q \pm 1)} T_{q \pm 1}^{(k)}\end{aligned}$$

Wigner-Eckart theorem:

$$\langle j' m' | T_q^{(k)} | j m \rangle = (-1)^{j' - m'} \begin{pmatrix} j' & k & j \\ -m' & q & m \end{pmatrix} \langle j' || \hat{T}^{(k)} || j \rangle$$

## Mixed tensor operators

$$\hat{Q}^{(k)} = \{\hat{T}^{(k_1)} \times \hat{U}^{(k_2)}\}^{(k)}$$

$$Q_q^{(k)} = \sum_{q_1, q_2} T_{q_1}^{(k_1)} U_{q_2}^{(k_2)} \langle k_1, q_1, k_2, q_2 | k_1, k_2, k, q \rangle$$

$$\begin{aligned} \langle l' || Q_q^{(k)} || l \rangle &= (-1)^{l'+k+l} \sqrt{2k+1} \\ &\times \sum_{l''} \left\{ \begin{matrix} k_1 & k & k_2 \\ l & l'' & l' \end{matrix} \right\} \langle l' || \hat{T}^{(k_1)} || l'' \rangle \langle l'' || \hat{U}^{(k_2)} || l \rangle \end{aligned}$$

Judd 1963: (3-39)

## Scalar product

Scalar product:

$$\begin{aligned}(\hat{T}^{(k)} \cdot \hat{U}^{(k)}) &= (-1)^k \sqrt{2k+1} \{\hat{T}^{(k)} \times \hat{U}^{(k)}\}_0^{(0)} \\ &= \sum_q (-1)^q T_q^{(k)} U_{-q}^{(k)}\end{aligned}$$

Generalisation to three tensors:

$$(\hat{T}^{(k)} \cdot \hat{U}^{(k')} \cdot \hat{V}^{(k'')}) = \sum_{q,q',q''} \begin{pmatrix} k & k' & k'' \\ q & q' & q'' \end{pmatrix} T_q^{(k)} U_{q'}^{(k')} V_{q''}^{(k'')}$$

## Recoupling of mixed tensor operators

$$\{\hat{T}_2^{(k_2)} \times \hat{T}_1^{(k_1)}\}^{(k)} = (-1)^{k_1-k_2-k} \{\hat{T}_1^{(k_1)} \times \hat{T}_2^{(k_2)}\}^{(k)}$$

$$\{\{\hat{T}_1^{(k_1)} \times \hat{T}_2^{(k_2)}\}^{(k_{12})} \times \hat{T}_3^{(k_3)}\}^{(k)} =$$

$$\sum_{k_{23}} \langle (k_1 k_2) k_{12}, k_3, k | k_1, (k_2 k_3) k_{23}, k \rangle \{\hat{T}_1^{(k_1)} \times \{\hat{T}_2^{(k_2)} \times \hat{T}_3^{(k_3)}\}^{(k_{23})}\}^{(k)}$$

Special cases:

$$\{\{\hat{T}_1^{(k_1)} \times \hat{T}_2^{(k_2)}\}^{(k_3)} \times \hat{T}_3^{(k_3)}\}^{(0)} = \{\hat{T}_1^{(k_1)} \times \{\hat{T}_2^{(k_2)} \times \hat{T}_3^{(k_3)}\}^{(k_1)}\}^{(0)}$$

$$\left( \{\hat{T}_1^{(k_1)} \times \hat{T}_2^{(k_2)}\}^{(k_3)} \cdot \hat{T}_3^{(k_3)} \right) =$$

$$(-1)^{k_3-k_1} \sqrt{\frac{2k_3+1}{2k_1+1}} \left( \hat{T}_1^{(k_1)} \cdot \{\hat{T}_2^{(k_2)} \times \hat{T}_3^{(k_3)}\}^{(k_1)} \right)$$

## Vector operators

Vector operators may be expressed as tensor operators of rank 1. The statement  $\vec{t} = \hat{t}^{(1)}$  implies the following rules:

$$t_x = \frac{1}{\sqrt{2}} (t_{-1}^{(1)} - t_{+1}^{(1)})$$

$$t_y = \frac{i}{\sqrt{2}} (t_{-1}^{(1)} + t_{+1}^{(1)})$$

$$t_z = t_0^{(1)}$$

This definition keeps the scalar product constant:

$$(\vec{t} \cdot \vec{v}) = (\hat{t}^{(1)} \cdot \hat{v}^{(1)})$$

## N-electron systems

Determinantal product states for  $\alpha = (nlm_l m_s)$ :

$$|\Psi\rangle = |\{\alpha_1 \alpha_2 \dots \alpha_N\}\rangle$$

General one-, two-, and three-electron N-particle operators as sums of elementary one-, two-, and three-electron operators.

$$F = \sum_i f_i \quad G = \sum_{i < j} g_{ij} \quad H = \sum_{i < j < k} h_{ijk}$$

Special case: The scalar product of two one-electron tensor operators splits into an one-electron and a two-electron part:

$$(\hat{U}^{(k)} \cdot \hat{T}^{(k)}) = \sum_i (\hat{u}_i^{(k)} \cdot \hat{t}_i^{(k)}) + \sum_{i < j} (\hat{u}_i^{(k)} \cdot \hat{t}_j^{(k)}) + \sum_{i < j} (\hat{u}_j^{(k)} \cdot \hat{t}_i^{(k)})$$

## Unit tensor operators

Unit tensor operator in orbital space:

$$\hat{U}^{(k)} = \sum_i \hat{u}_i^{(k)}$$
$$\langle l' || \hat{u}^{(k)} || l \rangle = \delta(l', l)$$

$$u_q^{(k)} |l, m\rangle = \sum_{m'} (-1)^{l-m'} \begin{pmatrix} l & k & l \\ -m' & q & m \end{pmatrix} |l, m'\rangle$$

Unit tensor operator in spin space:

$$\hat{T}^{(\kappa)} = \sum_i \hat{t}_i^{(\kappa)}$$
$$\langle s || \hat{t}^{(\kappa)} || s \rangle = 1$$

$$t_q^{(\kappa)} |m_s\rangle = \sum_{m'_s} (-1)^{s-m'_s} \begin{pmatrix} s & \kappa & s \\ -m'_s & q & m_s \end{pmatrix} |m'_s\rangle$$

## Reduction of two unit tensor operators

$$\{\hat{u}_i^{(k_1)} \times \hat{u}_i^{(k_2)}\}^{(k)} = (-1)^{2l+k} \sqrt{2k+1} \begin{Bmatrix} k_1 & k & k_2 \\ l & l & l \end{Bmatrix} \hat{u}_i^{(k)}$$

$$(\hat{u}_i^{(k)} \cdot \hat{u}_i^{(k)}) = \frac{(-1)^k}{\sqrt{(2l+1)(2k+1)}} \hat{u}_i^{(0)}$$

Special cases:

$$\{\hat{u}_i^{(1)} \times \hat{u}_i^{(k)}\}^{(k-1)} = -\frac{1}{2} \sqrt{\frac{(2l+k+1)(2l-k+1)k}{l(l+1)(2l+1)(2k+1)}} \hat{u}_i^{(k-1)}$$

$$\{\hat{u}_i^{(1)} \times \hat{u}_i^{(k)}\}^{(k)} = -\frac{1}{2} \sqrt{\frac{k(k+1)}{l(l+1)(2l+1)}} \hat{u}_i^{(k)}$$

$$\{\hat{u}_i^{(1)} \times \hat{u}_i^{(k)}\}^{(k+1)} = \frac{1}{2} \sqrt{\frac{(2l+k+2)(2l-k)(k+1)}{l(l+1)(2l+1)(2k+1)}} \hat{u}_i^{(k+1)}$$

$\hat{c}^{(k)}$  operator

$$\hat{C}^{(k)} = \sum_i \hat{c}_i^{(k)}$$

$$\langle l' || \hat{c}^{(k)} || l \rangle = (-1)^{l'} \sqrt{(2l' + 1)(2l + 1)} \begin{pmatrix} l' & k & l \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{aligned} c_q^{(k)} |lm\rangle &= \sum_{l', m'} (-1)^{m'} \sqrt{(2l' + 1)(2l + 1)} \\ &\quad \times \begin{pmatrix} l' & k & l \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l' & k & l \\ -m' & q & m \end{pmatrix} |l'm'\rangle \end{aligned}$$

## Reduction of two $\hat{c}^{(k)}$ operators

$$\{\hat{c}_i^{(k_1)} \times \hat{c}_i^{(k_2)}\}^{(k)} = (-1)^k \sqrt{2k+1} \begin{pmatrix} k_1 & k & k_2 \\ 0 & 0 & 0 \end{pmatrix} \hat{c}_i^{(k)}$$

Judd 1963: (4-5)

$$(\hat{c}_i^{(k)} \cdot \hat{c}_i^{(k)}) = (-1)^k \sqrt{2k+1} \hat{c}_i^{(0)}$$

Special cases:

$$\{\hat{c}_i^{(1)} \times \hat{c}_i^{(k)}\}^{(k-1)} = -\sqrt{\frac{k}{2k+1}} \hat{c}_i^{(k-1)}$$

$$\{\hat{c}_i^{(1)} \times \hat{c}_i^{(k)}\}^{(k)} = 0$$

$$\{\hat{c}_i^{(1)} \times \hat{c}_i^{(k)}\}^{(k+1)} = \sqrt{\frac{k+1}{2k+1}} \hat{c}_i^{(k+1)}$$

Judd 1963: (4-6, 4-7)

## Important one-electron tensor operators

$$\vec{L} = \sqrt{l(l+1)(2l+1)} \hat{U}^{(1)}$$

$$\vec{S} = \sqrt{3/2} \hat{T}^{(1)}$$

$$\vec{J} = \sqrt{l(l+1)(2l+1)} \hat{U}^{(1)} + \sqrt{3/2} \hat{T}^{(1)}$$

$$\hat{V}^{(k)} = \sqrt{2l+1} \hat{U}^{(k)}$$

## Important scalar product operators

$$\vec{L}^2 = l(l+1)(2l+1) (\hat{U}^{(1)} \cdot \hat{U}^{(1)})$$

$$\vec{S}^2 = 3/2 (\hat{T}^{(1)} \cdot \hat{T}^{(1)})$$

$$\begin{aligned} \vec{J}^2 = & l(l+1)(2l+1) (\hat{U}^{(1)} \cdot \hat{U}^{(1)}) + 3/2 (\hat{T}^{(1)} \cdot \hat{T}^{(1)}) \\ & + \sqrt{6l(l+1)(2l+1)} (\hat{U}^{(1)} \cdot \hat{T}^{(1)}) \end{aligned}$$

$$\hat{G}(R_{2l+1}) = \frac{1}{2l-1} \sum_{\substack{k=1 \\ k \text{ odd}}}^{2l-1} (2k+1) (\hat{U}^{(k)} \cdot \hat{U}^{(k)})$$

$$\hat{G}(G_2) = \frac{1}{4} [3 (\hat{U}^{(1)} \cdot \hat{U}^{(1)}) + 11 (\hat{U}^{(5)} \cdot \hat{U}^{(5)})]$$

## Important double tensor operators

$$\hat{V}^{(\kappa k)} = \sqrt{3/2} \hat{T}^{(\kappa)} \hat{U}^{(k)}$$

$$\hat{W}^{(\kappa k)} = \sqrt{(2k+1)(2\kappa+1)} \hat{T}^{(\kappa)} \hat{U}^{(k)}$$

$$\begin{aligned} \{\hat{W}^{(\kappa_1 k_1)} \times \hat{W}^{(\kappa_2 k_2)}\}^{(kk)0} = \\ \sqrt{(2k_1+1)(2k_2+1)(2\kappa_1+1)(2\kappa_2+1)} \\ \times \left\{ \{\hat{T}^{(\kappa_1)} \times \hat{T}^{(\kappa_2)}\}^{(k)} \times \{\hat{U}^{(k_1)} \times \hat{U}^{(k_2)}\}^{(k)} \right\}^{(0)} \end{aligned}$$

## Matrix elements of one-electron operators

If the set  $(\alpha'_1 \alpha'_2 \dots \alpha'_N)$  can be ordered so that  $\alpha'_i = \alpha_i$

- for all  $i$ , then

$$\langle \Psi' | F | \Psi \rangle = \sum_i \langle \alpha_i | f_1 | \alpha_i \rangle$$

- for all  $i$  except of  $i = N$ , then

$$\langle \Psi' | F | \Psi \rangle = \langle \alpha'_N | f_1 | \alpha_N \rangle$$

In all other cases the matrix element is zero.

## Matrix elements of two-electron operators

If the set  $(\alpha'_1 \alpha'_2 \dots \alpha'_N)$  can be ordered so that  $\alpha'_i = \alpha_i$

- for all  $i$ , then

$$\langle \Psi' | G | \Psi \rangle = \sum_{i < j} \langle \{ \alpha_i \alpha_j \} | g_{12} | \{ \alpha_i \alpha_j \} \rangle$$

- for all  $i$  except of  $i = N$ , then

$$\langle \Psi' | G | \Psi \rangle = \sum_i \langle \{ \alpha_i \alpha'_N \} | g_{12} | \{ \alpha_i \alpha_N \} \rangle$$

- for all  $i$  except of  $i = N - 1, N$ , then

$$\langle \Psi' | G | \Psi \rangle = \langle \{ \alpha'_{N-1} \alpha'_N \} | g_{12} | \{ \alpha_{N-1} \alpha_N \} \rangle$$

In all other cases the matrix element is zero.

## Matrix elements of three-electron operators

If the set  $(\alpha'_1 \alpha'_2 \dots \alpha'_N)$  can be ordered so that  $\alpha'_i = \alpha_i$

- for all  $i$ , then

$$\langle \Psi' | H | \Psi \rangle = \sum_{i < j < k} \langle \{\alpha_i \alpha_j \alpha_k\} | h_{123} | \{\alpha_i \alpha_j \alpha_k\} \rangle$$

- for all  $i$  except of  $i = N$ , then

$$\langle \Psi' | H | \Psi \rangle = \sum_{i < j} \langle \{\alpha_i \alpha_j \alpha'_N\} | h_{123} | \{\alpha_i \alpha_j \alpha_N\} \rangle$$

- for all  $i$  except of  $i = N - 1, N$ , then

$$\langle \Psi' | H | \Psi \rangle = \sum_i \langle \{\alpha_i \alpha'_{N-1} \alpha'_N\} | h_{123} | \{\alpha_i \alpha_{N-1} \alpha_N\} \rangle$$

- for all  $i$  except of  $i = N - 2, N - 1, N$ , then

$$\langle \Psi' | H | \Psi \rangle = \langle \alpha'_{N-2} \alpha'_{N-1} \alpha'_N | h_{123} | \alpha_{N-2} \alpha_{N-1} \alpha_N \rangle$$

In all other cases the matrix element is zero.

## Matrix elements of the unit tensor operator I

$$\langle l, m_{l_a} | u_q^{(k)} | l, m_{l_b} \rangle = (-1)^{l-m_{l_a}} \begin{pmatrix} l & k & l \\ -m_{l_a} & q & m_{l_b} \end{pmatrix}$$

Special case:

$$\langle l, m_{l_a} | u_0^{(0)} | l, m_{l_b} \rangle = \frac{\delta(m_{l_a}, m_{l_b})}{\sqrt{2l+1}}$$

## Matrix elements of the unit tensor operator II

$$\begin{aligned} \langle l, m_{l_a} | \{ \hat{u}^{(k_1)} \times \hat{u}^{(k_2)} \}_q^{(k)} | l, m_{l_b} \rangle = \\ (-1)^{2l+k} \sqrt{2k+1} \begin{Bmatrix} k_1 & k & k_2 \\ l & l & l \end{Bmatrix} \langle l, m_{l_a} | u_q^{(k)} | l, m_{l_b} \rangle \end{aligned}$$

Special case:

$$\langle l, m_{l_a} | (\hat{u}^{(k)} \cdot \hat{u}^{(k)}) | l, m_{l_b} \rangle = \frac{\delta(m_{l_a}, m_{l_b})}{2l+1}$$

## Matrix elements of the unit tensor operator III

$$\begin{aligned}
 \langle l, m_{l_a}, m_{l_b} | \{ \hat{u}_1^{(k_1)} \times \hat{u}_2^{(k_2)} \}_q^{(k)} | l, m_{l_c}, m_{l_d} \rangle = \\
 \delta(q_1, m_{l_a} - m_{l_c}) \delta(q_2, m_{l_b} - m_{l_d}) (-1)^{k_1 - k_2 + q} \sqrt{2k + 1} \\
 \times \begin{pmatrix} k_1 & k_2 & k \\ q_1 & q_2 & -q \end{pmatrix} \langle l, m_{l_a} | u_{q_1}^{(k_1)} | l, m_{l_c} \rangle \langle l, m_{l_b} | u_{q_2}^{(k_2)} | l, m_{l_d} \rangle
 \end{aligned}$$

Special case:

$$\begin{aligned}
 \langle l, m_{l_a}, m_{l_b} | (\hat{u}_1^{(k)} \cdot \hat{u}_2^{(k)}) | l, m_{l_c}, m_{l_d} \rangle = \delta(m_{l_a} + m_{l_b}, m_{l_c} + m_{l_d}) \\
 \times (-1)^{2l - m_{l_a} - m_{l_d}} \begin{pmatrix} l & k & l \\ -m_{l_a} & m_{l_a} - m_{l_c} & m_{l_c} \end{pmatrix} \begin{pmatrix} l & k & l \\ -m_{l_b} & m_{l_b} - m_{l_d} & m_{l_d} \end{pmatrix}
 \end{aligned}$$